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Knowledge-based recommendation: a review of ontology-based recommender systems for e-learning

John K. Tarus^{1,2} · Zhendong Niu^{1,3} · Ghulam Mustafa⁴

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Abstract Recommender systems in e-learning domain play an important role in assisting the learners to find useful and relevant learning materials that meet their learning needs. Personalized intelligent agents and recommender systems have been widely accepted as solutions towards overcoming information retrieval challenges by learners arising from information overload. Use of ontology for knowledge representation in knowledge-based recommender systems for e-learning has become an interesting research area. In knowledge-based recommendation for e-learning resources, ontology is used to represent knowledge about the learner and learning resources. Although a number of review studies have been carried out in the area of recommender systems, there are still gaps and deficiencies in the comprehensive literature review and survey in the specific area of ontology-based recommendation for e-learning. In this paper, we present a review of literature on ontology-based recommenders for e-learning. First, we analyze and classify the journal papers that were published from 2005 to 2014 in the field of ontology-based recommendation for e-learning. Secondly, we categorize the different recommendation techniques used by ontology-based e-learning recommenders. Thirdly, we categorize the knowledge representation technique, ontology type and ontology representation language used by ontology-based recommender systems, as well as types of learning resources recommended by e-learning recommenders. Lastly, we discuss the future trends of this recommendation approach in the context of e-learning. This study shows that use of ontology for knowledge representation in e-learning recommender systems can improve the quality of recommendations. It was also evident that hybridization of knowledge-based

✉ Zhendong Niu
zniu@bit.edu.cn

John K. Tarus
johnktarus@yahoo.com

¹ School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China

² Directorate of ICT, Moi University, Eldoret, Kenya

³ School of Information Sciences, University of Pittsburgh, Pittsburgh, PA, USA

⁴ Department of Information Technology, University of the Punjab, Jhelum, Pakistan

recommendation with other recommendation techniques can enhance the effectiveness of e-learning recommenders.

Keywords E-learning · Recommender systems · Ontology · Knowledge-based · Recommendation

1 Introduction

As online learning resources increase exponentially on the World Wide Web, online learners on the other hand experience difficulty in choosing the most suitable and relevant learning materials that meet their learning needs due to information overload. With the advent of the Internet, choosing information from numerous alternatives has been a great challenge due to information overload (Montaner et al. 2003). Recommender systems can help learners overcome this problem by automatically recommending the most relevant learning resources to the learners according to their personalized preferences and profile. Research on recommender systems for e-learning has become an interesting area for researchers in this field. A recommender system has been defined as software tools and techniques providing suggestions for items to be of use to a user (Ricci et al. 2011). The idea of personalized and intelligent agents, search engines and recommender systems has been widely accepted as solutions towards overcoming information retrieval challenges arising from information overload (Montaner et al. 2003). However, unlike the search engines or retrieval systems that return relevant results that match the user query, recommender systems suggest personalized recommendations tailored to the user needs and preferences. They play an important role in areas such as e-commerce and e-learning (Pan et al. 2010). Recommendations for books on Amazon, or movies on Netflix, are examples of real-world recommender systems.

Moreover, study of recommender systems for e-learning domain has become an important research field (Manouselis et al. 2011). These systems help the learners in e-learning environment to quickly find relevant learning resources that are suitable for their learning needs. One of the challenges of this type of system is to perform a proper and accurate combination between the users' expectations (their profiles) and the items to be recommended (Cazella et al. 2014). The requirements for recommenders in the e-learning domain are different from other domains. Researchers have been working on recommender systems using several techniques but with the same goal of filtering out irrelevant information from relevant information (Cheng et al. 2013).

In the context of e-learning, different learners have different characteristics in terms of learners' background knowledge, learners' history, competence level, learning style and learning activities (Buder and Schwind 2012) which makes recommendation of learning resources to a particular learner more difficult. As such, even if two learners have similar ratings, they will require different recommendations if the learner characteristics are not the same. Klačnja-Milićević et al. (2011) point out that recommender systems for e-learning need to consider the learner's specific demands and requirements. It is therefore increasingly important to consider these characteristics for a particular learner in the personalization and recommendation process in e-learning environments. Conventional recommender systems like collaborative filtering (CF) and content-based (CB) recommenders do not take these learner characteristics into consideration in their recommendation process, instead, they recommend items to the users based solely on ratings. This complexity in recommendation of learning resources can be alleviated by personalizing the learner profile to match the needs

and characteristics of the learner by using knowledge structures such as ontology. [Verbert et al. \(2012\)](#) emphasize the importance of incorporation of additional information about the learner, teacher and their context in the recommendation process.

Aggregation of ontology domain knowledge into the recommendation process is one of the solutions that can overcome the limitations of conventional recommender systems. Ontology-based (OB) recommenders are knowledge-based recommender systems which use ontology to represent knowledge about the items and users in the recommendation process. In e-learning, ontology-based recommender systems use ontology knowledge about the learner and learning resources in mapping a learner to relevant learning resources. Ontologies play an important role in knowledge representation as well as in knowledge sharing and reuse in these systems. Previous studies have shown that aggregation of ontology domain knowledge about the learner and learning resources improves the accuracy and quality of recommendations as well as alleviate other drawbacks associated with conventional recommendation techniques such as cold-start and rating sparsity problems ([Adomavicius and Tuzhilin 2005](#); [Zhao et al. 2015b](#)).

In this work, we present a comprehensive review of publications on ontology-based recommender systems for e-learning from 2005 to 2014. The review is guided by the guidelines for conducting a systematic literature review proposed by [Kitchenham and Charters \(2007\)](#) which have been used by other authors in conducting similar reviews such as [Najafabadi and Mahrin \(2016\)](#) and [González-Martínez et al. \(2015\)](#). Our survey makes three major research contributions that distinguish it from other related studies. The main contributions of this work are:

- (1) First, we summarize the research achievements in the class of ontology-based recommendation from 2005 to 2014. Additionally, we have graphically illustrated the distribution of the journal paper publications per year during the same period.
- (2) Secondly, we analyze and categorize the different recommendation and knowledge representation techniques, ontology types, as well as ontology representation languages used by ontology-based recommenders for e-learning. We also identify the kinds of learning resources recommended by ontology-based recommenders for e-learning.
- (3) Lastly, we discuss and give an insight on the future research trends on this area of recommender systems.

Although a number of review studies have been carried out on other areas of recommender systems, no survey has been carried out on ontology-based recommender systems for e-learning. Most of the previous review studies focused more on conventional recommendation techniques like collaborative filtering and content-based recommendation ([Bobadilla et al. 2013](#); [Manouselis et al. 2011](#)), hybrid recommendation ([Burke 2007](#)) and other emerging techniques such as context-aware (CA) recommenders ([Abbas et al. 2015](#); [Verbert et al. 2012](#)). A more recent review on recommender systems in e-learning environments by [Klašnja-Milićević et al. \(2015\)](#) focused on tag-based recommender systems. The novelty of our survey focuses specifically on ontology-based recommenders for e-learning of which to the best of our knowledge, no review study has been conducted on this area.

The rest of this paper is structured as follows. In Sect. 2, we review the related work relevant to this study including recommendation techniques and previous review studies. In Sect. 3, we briefly discuss the background on ontology-based recommenders for e-learning. In Sect. 4, we explain the methodology adopted in this review study. In Sect. 5, we present our results and discussion including the future trends on ontology-based recommendation and finally in Sect. 6, we conclude this study.

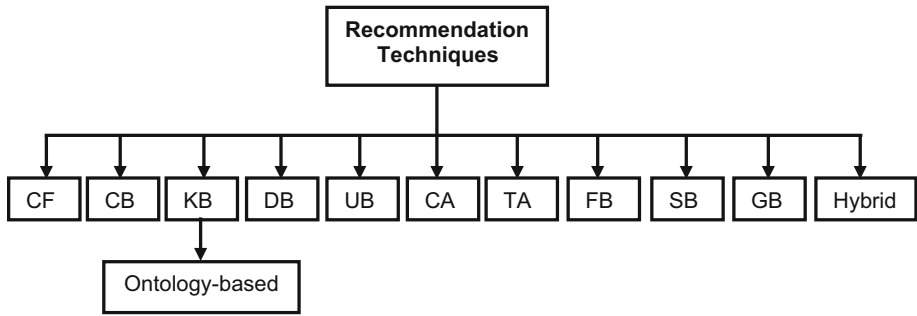


Fig. 1 Recommendation techniques

2 Related work

This section presents a review of the existing literature relevant to this study with a focus on a brief overview of common recommendation techniques and previous related review studies.

2.1 Common recommendation techniques

Recommender systems have become more popular in the field of e-learning as a way of solving the problem of retrieval of relevant and useful learning materials by learners arising from information explosion on the internet. Recommender systems are classified according to the technique used in recommendation. In this sub-section, we present a brief overview of the different recommendation techniques. [Burke \(2007\)](#) and [Adomavicius and Tuzhilin \(2005\)](#) distinguish between different classes of recommendation techniques which include collaborative filtering (CF), content-based (CB), knowledge-based (KB), demographic-based (DB), utility-based (UB) and hybrid recommendation. Other advanced recommendation techniques include context-aware (CA) ([Adomavicius and Tuzhilin 2011](#); [Huang et al. 2011b](#)), trust-aware based (TA), fuzzy-based (FB) ([Zhang et al. 2013b](#)), social network-based (SB) ([He and Chu 2010](#)), group-based (GB) ([Masthoff 2011](#)) and ontology-based techniques. Each approach has its strengths as well as limitations. Ontology-based recommenders are knowledge-based recommender systems that use ontology for knowledge representation ([Yang 2010](#)). Figure 1 illustrates the taxonomy of recommendation techniques.

Collaborative filtering (CF) Recommends to the active user the items that other users with similar tastes liked in the past. The similarity in taste of two users is calculated based on the similarity in the rating history of the users ([Schafer et al. 2007](#)). A rating measures the degree of interest in an item by a user. Computation of similarity between users is the principle behind collaborative filtering. The two most popular similarity measures are correlation-based and cosine-based similarities ([Sharma and Mann 2013](#)). [Chen et al. \(2014\)](#) singles out Adjusted Cosine Similarity (Eq. 1) as the most commonly used measurement of similarity. Similarity between two items i and j is calculated as follows:

$$\text{sim}(i, j) = \frac{\sum (r_{u,i} - \bar{r}_u)(r_{u,j} - \bar{r}_u)}{\sqrt{\sum (r_{u,i} - \bar{r}_u)^2} \sqrt{\sum (r_{u,j} - \bar{r}_u)^2}} \quad (1)$$

where $r_{u,i}$ is the rating given to item i by user u , $\bar{r}_u$ is the mean rating of all the ratings provided by u ([Jannach et al. 2011](#), p. 19).

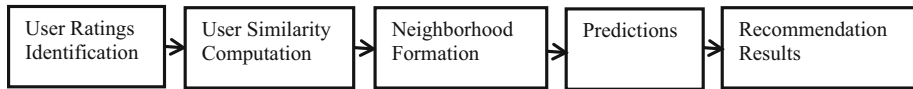


Fig. 2 Recommendation process in collaborative filtering

The most widely used algorithm for collaborative filtering is the k Nearest Neighbors (k NN) (Adomavicius and Tuzhilin 2005; Schafer et al. 2007; Bobadilla et al. 2011). Nearest neighbors are users with similar tastes with the active user in terms of ratings similarity. In the user to user version, k NN executes the following three tasks to generate recommendations for an active user: (1) determine k users neighbors (neighborhood) for the active user a ; (2) implement an aggregation approach with the ratings for the neighborhood in items not rated by a ; and (3) extract the predictions in step 2 then select the top k recommendations (Bobadilla et al. 2013). Figure 2 summarizes the entire process of recommendation in collaborative filtering.

Though collaborative filtering is the most popular recommendation technique (Ricci et al. 2011), its major drawback is new user and new item problems as well as the rating sparsity problem. New user problem occurs in scenarios where a new user who has recently entered the system has not rated any items, hence cannot receive recommendations due to lack of user preferences while a new item problem occurs in the event a new item has too few ratings, hence cannot be recommended. The new user and new item problems are commonly referred to as cold-start problem which occurs in scenarios where it is not possible to make reliable recommendations due to an initial lack of ratings (Adomavicius and Tuzhilin 2005; Schafer et al. 2007). Rating sparsity occurs when few users have rated the same item, hence no overlap in rating preferences with the target user (Zhao et al. 2015b).

Content-based (CB) In content-based recommendation, the system learns to recommend items that are similar in terms of content features to the ones that the user liked in the past. The similarity of items is calculated based on the features associated with the compared items (Burke 2007; Pazzani and Billsus 2007). Computation of similarity between items is therefore the principle behind content-based recommendation. Content based recommenders can be classified into case-based reasoning (Adomavicius and Tuzhilin 2005) and attribute-based technique (Drachsler et al. 2008). Case based reasoning technique recommend items with the highest correlation to items the learner liked before while attribute-based technique recommend items based on the matching of their attributes to the learner profile (Klašnja-Milićević et al. 2015).

Most content-based recommender systems use relatively simple retrieval models, such as keyword matching or Vector Space Model (VSM) with basic Term Frequency-Inverse Document Frequency (TF-IDF) weighting. TF-IDF is used to compute the overall importance of keywords in a document (Eq. 2) (Lops et al. 2011, p. 82).

$$TF_IDF(t_k, d_j) = \underbrace{TF(t_k, d_j)}_{TF} \bullet \underbrace{\log \frac{N}{n_k}}_{IDF} \quad (2)$$

where d_j denotes the document in a n -dimensional vector space ($d_j = \{w_{1j}, w_{2j}, \dots, d_{nj}\}$), t_k denotes the term (keyword) in document d_j , N denotes the number of documents in the corpus, and n_k denotes the number of documents in the collection in which the term t_k occurs at least once.

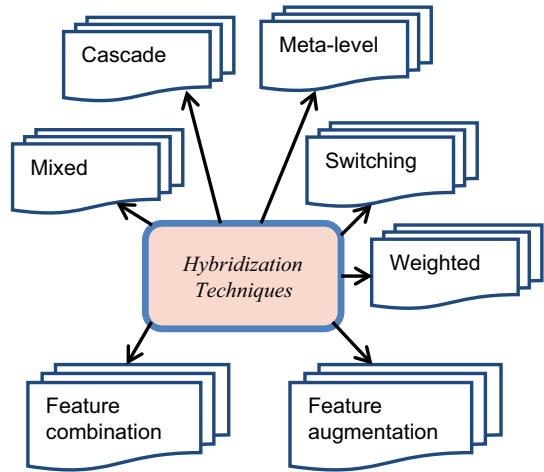
The main advantages of content-based recommenders are user independence, transparency, and new unrated item recommendation. However, it experiences some drawbacks such as limited content analysis, overspecialization, serendipity and new user problem (Ricci et al. 2011). Limited content analysis refers to scenarios where there is difficulty in extraction of useful and reliable information from heterogeneous data formats like video, audio and images and this in turn affects the quality of recommendation results. Overspecialization occurs in situations where a user receives recommendations only from items very similar to their preferences or what they liked before, hence the users are unable to receive recommendations of items that they do not know but are likely to be of interest to them. Serendipity refers to making discoveries of items by accident which were not in quest of during recommendation. Content-based recommender systems need to accumulate enough ratings before overcoming the startup problem.

Knowledge-based (KB) recommenders These systems recommend items to users based on domain knowledge about how items meet user preferences (Colombo-Mendoza et al. 2015). Knowledge-based recommender systems need to employ three types of knowledge; knowledge about the users, knowledge about the items and knowledge about the matching between the item and user's need (Adomavicius and Tuzhilin 2005). In the context of e-learning, knowledge-based technique aggregates knowledge about the learners and learning materials to apply them in the recommendation process (Shishehchi et al. 2012). Unlike collaborative filtering and content-based, knowledge-based recommenders don't experience ramp-up/cold-start problem (Adomavicius and Tuzhilin 2005) or rating sparsity problem (Zhao et al. 2015b) since their recommendations are not tied to ratings but instead, they make use of the domain knowledge. Knowledge-based techniques are therefore suitable for hybridization with other recommendation techniques in the case of e-learning recommenders. However, the main drawback of knowledge-based recommenders is requirement of knowledge engineering skills (Burke 2007).

Ontology-based (OB) recommenders Ontology is an explicit formal specification of a shared conceptualization (Gruber 1993). Ontologies are used to model knowledge about the user context, knowledge about the item, and the domain knowledge (Yu et al. 2007). In ontology-based e-learning recommenders, ontology is used for knowledge representation (Shishehchi et al. 2012). Ontology-based recommenders are knowledge-based recommender systems that use ontology for knowledge representation. In the context of e-learning domain, ontology is used to model the knowledge about the learner and learning resources. Like knowledge-based recommender systems, ontology-based recommenders do not experience most of the problems associated with conventional recommender systems such as cold-start problem, rating sparsity (Zhao et al. 2015b) and overspecialization since ontology-based recommenders rely more on domain knowledge rather than ratings. This feature makes ontology-based recommenders suitable for e-learning domain (Zhang et al. 2013a; Yang 2010). On the other hand, construction of ontologies is a difficult and expensive process. Furthermore, ontology creation is time consuming. To overcome these challenges, modern tools for automatic creation of ontologies have being proposed (Santos and Boticario 2015).

Hybrid recommender systems The hybrid filtering method hybridizes the features of two or more recommendation techniques e.g. content-based filtering and collaborative filtering systems to benefit from the strengths of each technique and to improve performance (Ghauth and Abdullah 2010). Hybrid recommendation technique is very useful because it can overcome most of the limitations experienced by the individual conventional recommendation approaches by combining them for better results. For example, Chen et al. (2014) propose a

Fig. 3 Hybridization techniques



hybrid learning resource recommendation algorithm which combines collaborative filtering and sequential pattern mining for recommending learning materials. Similarly, [Zhao et al. \(2015a\)](#) propose a hybrid approach which uses topic model in the first step and matrix factorization in the second to solve the recommendation problem. The results in both cases showed improved performance of their recommender systems. There are many methods of hybridizing recommendation techniques ([Burke 2007](#), pp. 377–408) as shown in Fig. 3.

Demographic-based (DB) and Utility-based (UB) recommender systems Demographic-based recommender systems aim to categorize the user based on personal attributes and make recommendations based on demographic classes. Users with common or similar personal attributes are likely to have similar preferences of items. On the other hand, utility-based recommenders make suggestions based on a computation of the utility of each object for the user ([Burke 2007](#); [Pazzani 1999](#)).

Context aware-based (CA) recommender systems [Dey et al. \(2001\)](#) define context as any information that can be used to characterize the situation of an entity. In context-aware approach, contextual information is used to filter the data set before applying a traditional recommendation algorithm ([Adomavicius and Tuzhilin 2011](#); [Verbert et al. 2012](#); [Huang et al. 2011b](#)).

Trust-aware based (TA) recommender systems [Golbeck and Hendler \(2006\)](#) define trust in a person as a commitment to an action based on a belief that the future actions of that person will lead to a good outcome. The goal of trust-based recommendation is to generate personalized recommendations from known opinions and trust relationships. In e-commerce and e-learning recommenders, trust is an important aspect to improve social network relationship among users ([Moradi and Ahmadian 2015](#)).

Social-network recommender systems These recommender systems generate recommendations based on information about users' profiles and relationships between users ([Victor et al. 2011](#), pp. 645–675; [He and Chu 2010](#), pp. 47–74). Examples of sites using social-network recommendation are Facebook and LinkedIn.

Fuzzy-based recommender systems This refers to recommender systems where recommendations to users are generated on the basis of uncertain or vague information. For example,

in many practical situations, users express their preference for items in linguistic terms such as ‘very interested’ or ‘not interested’ (Zhang et al. 2013b).

Group recommender systems Group recommender systems recommend items to a group of users rather than to an individual. This kind of recommendation technique is suitable to situations where several people participate in a single activity (Masthoff 2011). For example, a recommender may select television programmes for a group to view.

2.2 Previous related review studies

A number of review papers have been published in the last few years on e-learning recommender systems. For instance, Verbert et al. (2012) conducted a survey of context-aware recommender systems that have been deployed in technology enhanced learning (TEL) settings. In their review, they describe a context framework that identified the relevant context dimensions such as time, location, activity, physical conditions, resource and social relations for TEL applications as well as an analysis of existing TEL recommender systems along the context dimensions. Similarly, Abbas et al. (2015) carried out a survey on context-aware recommender systems based on computational intelligence techniques. The key feature in their survey was the presentation of taxonomy of the computational intelligence techniques and challenges particular to the context-aware recommender systems. Erdt et al. (2015) reviewed technology enhanced learning (TEL) recommender systems by carrying out a quantitative analysis and classification of papers according to the following criteria: type of evaluation methodology, subject of evaluation, and effects measured by the evaluation. Yang et al. (2014) reviewed studies on collaborative filtering based social recommender systems and classified CF-based social recommender systems into two categories: matrix factorization based social recommendation approaches and neighborhood based social recommendation approaches. In Lu et al. (2015) review of recommender systems, they clustered their applications into eight categories and then summarized the related recommendation techniques used in each category. The categories included e-government, e-business, e-commerce/e-shopping, e-library, e-learning, e-tourism, e-resource services and e-group activities. In early work of Adomavicius and Tuzhilin (2005), they carried out a state of the art review of the field of recommender systems and described the conventional recommendation techniques namely content-based, collaborative, and hybrid recommendation. Their work focused on the limitations of the conventional approaches and proposed possible extensions that can address drawbacks such as understanding of users and items, incorporation of the contextual information into the recommendation process and support for multicriteria ratings.

Moreover, Park et al. (2012) carried out a review on recommender systems and classified them according to year of publication, journal in which they are published, the application field and data mining techniques. In Pu et al. (2012) review of recommender systems from the user’s perspective, they focused on three key interactions between the user and the system: the initial preference elicitation process, the preference refinement process, and the presentation of the system’s recommendation results. Drachsler et al. (2015) analyzed recommender systems in TEL from 2000 to 2014 and categorized 82 recommender systems into 7 clusters according to their characteristics and their contribution to research in TEL recommender systems. In their work, the 16 papers analyzed under cluster 3 focuses on educational knowledge as information source for the recommendation process as well as explicit description of knowledge in terms of rules, ontologies, concept maps and semantic relations. More recently, Klačnja-Milićević et al. (2015) conducted a comprehensive review of recommender systems in e-learning environments with a focus on important requirements and challenges in design-

ing recommender systems in e-learning environments. The paper further proposed possible extensions with model for tagging activities and tag-based recommender systems.

Literature on previous related reviews on recommender systems shows that a number of review studies on conventional recommendation approaches like collaborative filtering, content-based and hybrid recommendation as well as advanced techniques like context-aware techniques have been published. However, none of the existing papers on recommender systems survey addresses ontology-based recommendation for e-learning, and as a result, our review fills this gap.

3 Ontology-based (OB) recommenders in e-learning

Ontology is a formal explicit description of concepts in a domain consisting of objects, classes (concepts), properties, relationships, rules and constraints (Poelmans et al. 2013). Encoding of knowledge about specific domains is achieved using ontology representation languages such as Web Ontology Language (OWL), Resource Description Framework (RDF), DARPA Agent Markup Language (DAML) and Ontology Inference Layer (OIL) (Poelmans et al. 2013). OWL and RDF are the commonly used formal languages for constructing ontologies. Moreover, ontology representation languages can be used to formally describe concepts and properties between them (Ruotsalo 2010; Kalibatiene and Vasilecas 2011). These encoding languages often include predefined semantics and reasoning rules that support processing of that knowledge. An example of the rule language is the Semantic Web Rule Language (SWRL) which is a language specifically targeted to introduce inference rules in knowledge models represented in OWL (Sicilia et al. 2011). For instance, Vesin et al. (2012) ontology is encoded using OWL representation language with SWRL and Jess rules for reasoning. Kontopoulos et al. (2008) domain ontology is encoded with RDF representation language.

Ontologies can be classified based on the domain scope. The categories in this classification include application ontology, domain ontology, reference ontology, general ontology and top level/generic ontology (Roussey et al. 2011, pp. 9–38). Application ontologies are specializations of domain ontologies which represents the particular model of a domain according to a single viewpoint of a user or a developer. Domain ontologies represent knowledge about a particular part of the world. Reference ontologies on the other hand represent knowledge about a particular part of the world in a way that is independent from specific objectives, through a theory of the domain represented (Burgun 2006). General ontologies are not dedicated to a specific domain or field. They contain general knowledge of a huge area. Examples of general ontologies are CYC and DBpedia. CYC is a general knowledge base and commonsense reasoning engine while DBpedia represents a community effort to structure information from Wikipedia and make this information available on the web (Paredes-Valverde et al. 2015). Top level ontologies are generic ontologies applicable to various domains. They define basic notions like objects, relations, events, processes and so on (Roussey et al. 2011, pp. 9–38). Vesin et al. (2012) used two main kinds of ontologies namely domain ontology and task ontology while Cobos et al. (2013) recommender is based on a reference ontology.

Ontologies are used in different domains with different goals. For instance, recommender systems use domain ontology to establish the relationships between users and their preferences about the recommendation subject. In addition, they use ontologies to complement other techniques for knowledge representation (Martinez-Cruz et al. 2015). Most knowledge based e-learning recommender systems use ontologies to represent knowledge about the

learner and learning resources. In such a case, ontology is used to establish the relationship between learners and their preferences about the learning resources. Those knowledge based recommenders that use ontologies for knowledge representation are called ontology-based recommender systems (Ruotsalo 2010). In Cheng et al. (2013) recommender system, ontology was used for personalization strategies whereas in Zhang et al. (2013a) ontology was used for user modeling.

The goal of e-learning recommender systems is to assist the learners to find relevant and useful learning materials that meet their learning needs. Whereas conventional recommender systems generally recommend items to the users based on ratings, e-learning recommender systems require additional information on learner characteristics such as learner history, knowledge level and learning style for personalization of the learner profile. Salehi et al. (2013) emphasizes that one of the most important applications of recommendation systems in an e-learning environment is personalization and recommendation of learning resources. Ontology-based recommendation is one of the techniques that are commonly used for recommending learning resources to learners in an e-learning environment. García et al. (2013) point out that ontologies have demonstrated to be the most convenient structure for representing knowledge models. Apart from ratings and user or item similarity (Zhao et al. 2015b), ontology-based recommenders in e-learning environments consider the knowledge about the learner and learning objects in the recommendation process. This is because unlike in other domains, recommender systems for e-learning domain should take into account additional information such as background knowledge, learner history and learning style of the learner during the recommendation process. To demonstrate this, Chen et al. (2014) point out that a user will probably learn from easy materials to difficult ones; for a single knowledge point, a user will probably learn from theoretical to practical, hence this historical learning sequence should be captured. Similarly, the heterogeneous formats of learning materials which include text, images, audio and video should be considered in choosing a recommendation technique. For example, Klačnja-Milićević et al. (2011) propose a personalized e-learning system which can automatically adapt to the interests, habits and knowledge levels of learners. Ontology-based recommenders aggregate the knowledge about the learner and learning materials to apply them in the recommendation process (Shishhehchi et al. 2012). One of the major strengths of ontology-based systems arising from use of domain knowledge is its ability to improve the quality of recommendations as well as overcome problems associated with conventional recommenders such as cold-start and rating sparsity problem.

The effectiveness of ontology-based recommender systems depends mainly on the completeness and accuracy of knowledge maintained in the ontology domain knowledge. The benefits of using ontologies for knowledge representation have been pointed out by Medland (2007).

A number of researchers have used ontology for knowledge representation in e-learning resource recommendation and their results have shown improvement in the quality of recommendations. For instance, Sosnovsky et al. (2012) used ontology in their personalized recommender system for learning material. Their approach which was implemented in an adaptive system recommends supplementary reading material and adaptively annotates it. Han et al. (2010) equally propose a new ontology-based learning object recommendation framework for semantic content recommendation towards learning process. Their recommender is made up of three components namely ontology, semantic rules and concept lattices. Ontology represents the learning object as well as modelling the cognitive structure of the learner; semantic rules identifies the prerequisite concepts contributing to the understanding of the current learning object; while concept lattices are for obtaining the final recommendations. OWL representation language was used for encoding the ontology knowledge. Other

authors including Pukkhem (2014), Vesin et al. (2012), Cobos et al. (2013) and Shishehchi et al. (2012) also used ontology for knowledge representation in their work. Representation of ontology in Pukkhem (2014) is based on OWL with Semantic Web Rule Language (SWRL) for reasoning rules. To encode knowledge, Cobos et al. (2013) used both OWL and RDF ontology representation language to construct reference ontology with 23 classes and 76 instances. The ontology described in Shishehchi et al. (2012) was encoded with OWL and had two main classes namely learner class and learning material class.

Ontology-based recommenders have gained popularity in e-learning resources recommendation in the last few years due to their ability to personalize learner profiles based on the characteristics of the learner such as background knowledge, learning style, learning paths and knowledge level.

4 Methodology

The methodological guidelines adopted in this work are those recommended by Kitchenham and Charters (2007) for performing systematic literature reviews in Software Engineering. The following subsections explain the methodology. First, the selection criteria of publications reviewed in the study is provided. Next is how the analysis, classification and categorization of the publications were conducted.

4.1 Selection of publications reviewed in this survey

The procedure involved searching and downloading journal papers from year 2005 to 2014 from different top level electronic journal databases which included Web of Science (SCI), Engineering Index (EI), Science Direct, EBSCO Academic Search Premier, Springer, IEEE Xplore and ACM Digital Library. The search and retrieval of publications was carried out in November 2015. These electronic journal databases were chosen because they contained published papers on the topic under review. Two sets of papers were downloaded and reviewed in this survey.

The first category of papers was those of previous related review studies in the area of recommender systems for e-learning as well as papers related to recommendation techniques. The purpose of this category of papers was to establish what review studies have been carried out by other authors previously as well as give a brief background of the field of recommender systems. Publications reviewed in this category included both journal and conference papers. A total of 80 papers were reviewed in this category for related work relevant to this study.

The second category consisted of papers specific to this particular survey on ontology-based recommenders for e-learning domain. In the second category of papers, only journal articles published from 2005 to 2014 were considered. Journal articles were considered because of the quality of their content since they undergo rigorous peer review and they are detailed. In searching the papers from the electronic journal databases in this category, the keywords “ontology”, “recommender”, “recommendation”, “learning”, “education”, “knowledge”, “e-learning” and “online learning” were among the main keywords used in filtering the search results. The Boolean operators “OR” and “AND” were used in formulating the different search string query which ensured a broad and exhaustive set of possibilities of the search are covered. The search string was formulated as: (“ontology” OR “knowledge” OR “knowledge-based” OR “ontology-based”) AND (“recommender” OR “recommendation”) AND (“learning” OR “education” OR “e-learning” OR “online learning”). The search

string query was then applied to all the prestigious electronic journal databases mentioned above.

After retrieval of the articles, the authors carried out the first review by reading through each paper with a focus on the title, keywords and abstract in order to preselect the articles relevant to this review. A second review of selected articles that involved reading the entire paper was then carried out and the papers that did not meet the selection criteria were excluded. This approach was also used previously in other review studies by [González-Martínez et al. \(2015\)](#), [Najafabadi and Mahrin \(2016\)](#) and [Hashizume et al. \(2013\)](#). The papers were selected based on their relevance to the area under study and if they met the following inclusion criteria: (1) has a distinct recommendation technique(s); (2) uses ontology as one of the techniques in knowledge representation; (3) published in a top level journal from 2005 to 2014 taking into account the impact factor of the journal where applicable; (4) the application domain area is e-learning.

However, those articles that were retrieved but did not meet the inclusion criteria were excluded. Such articles were deemed to contain insufficient information for the review to warrant inclusion. The exclusion criteria were:

- The recommendation and knowledge representation techniques used in the recommender system are not clearly indicated by the authors of the article.
- The domain of recommendation is not e-learning.
- The recommended items are neither learning resources nor learning activities.
- Articles in which only the abstract was available without the full paper.
- Secondary studies such as survey papers whose main goal was to report the results of review studies.

In the initial stage before exclusion, a total of 229 papers were retrieved. After applying the above-mentioned exclusion and inclusion selection criteria and performing some deep reading on the retrieved papers, 36 relevant papers were finally selected for intensive qualitative analysis and review in this study. These were the papers that were specific to the goals of this review study and met the selection criteria. The papers were then analyzed, classified and categorized. It should be noted that the 36 publications that met the selection criteria were limited to journal paper articles published from 2005 to 2014.

4.2 Analysis, classification and categorization of publications

The articles that met the selection criteria for category two articles were analyzed qualitatively using a content analysis technique then classified according to the year of publication and categorized. The main purpose of classification was to ease the process of analysis and graphically show the growth of the publications in this area from year 2005 to 2014. Another aim of classification was to graphically summarize the research achievements in this area of study.

Qualitative analysis and categorization of the selected publications relevant to ontology-based recommenders for e-learning domain was carried out by the three researchers with a view to extracting the information of interest in this review. Information of interest in the analysis involved identification of the author and year of publication, recommendation technique(s), knowledge representation technique(s), and recommended learning resources. This information of interest was extracted and categorized accordingly in tables. Data extracted from the relevant publications by each of the three authors was compared and any disagreements resolved by consensus as suggested in [Kitchenham and Charters \(2007\)](#). This data extraction approach was also used by [Zydney and Warner \(2016\)](#) in their review study.

5 Results and discussion

In this section, we present an overview of selected publications on ontology-based recommender systems and results of our survey on ontology-based (OB) recommender systems for e-learning. The results are presented in line with the objectives of this review study namely: classify the journal papers that were published from year 2005 to 2014 in the field of ontology-based recommendation for e-learning; categorize the different recommendation techniques used by ontology-based e-learning recommenders; categorize ways of knowledge representation, ontology types, ontology representation language as well as identify the types of learning resources recommended by e-learning recommenders. Lastly, discuss the future trends of this recommendation approach in the context of e-learning. Out of a total of 116 relevant publications selected according to the criteria in Sect. 4, 36 papers were directly relevant to ontology-based recommender systems for e-learning. The papers directly relevant to this survey were analyzed, classified and categorized as explained in the following sub-sections.

5.1 Overview of selected publications on ontology-based recommender systems for e-learning

This sub-section presents a brief overview of the selected 36 publications directly relevant to ontology-based recommenders for e-learning. A number of studies on the area of ontology-based recommendation in e-learning have been carried out from year 2005 to 2014 and as such, many researchers have attempted to aggregate ontology domain knowledge into the recommendation process for knowledge representation. For instance, [Capuano et al. \(2014\)](#) developed an adaptive e-learning recommender system named Intelligent Web Teacher (IWT) which uses a combination of collaborative filtering and ontology knowledge structures in its recommendation approach. Similarly, [Pukkhem \(2014\)](#) presents “LORecommendNet”, which uses ontology to represent knowledge about learning objects, learner model and semantic mapping rules. [Zuhadar and Nasraoui \(2010\)](#) propose a hybrid recommender system that uses multi-ontology models to recommend educational content in e-learning repository. [Shen and Shen \(2005\)](#) propose a recommender system that uses ontology for knowledge representation together with sequencing rules to make intelligent recommendations. [Zhang et al. \(2013a\)](#) propose an ontology-based collaborative filtering recommendation algorithm which could help users find the nearest neighbors and calculate the similarity of user rating knowledge points. On the other hand, [Yang \(2010\)](#) developed an ontology-supported recommendation system for scholars that can integrate specific documents as well as extract important information.

Furthermore, [Huang et al. \(2011a\)](#) propose personalized recommendation for learning paths, learning contents and learning experiences that uses ontologies for representing knowledge about the learners, course and learning objects. The results from their experiments show that their system can improve the learning performance. [Blanco-Fernández et al. \(2011\)](#) present a content-based recommendation strategy that uses techniques such as spreading activation and semantic associations as well as ontology for discovering additional knowledge about the learner preferences. In [Vesin et al. \(2011\)](#), PROgramming TUtoring System—Protus is an intelligent ontology-based recommender system for learning essential concepts of Java programming language. [Brut and Sèdes \(2010\)](#) propose a knowledge-based personalized recommender system for e-learning materials. Their system uses ontology for knowledge representation. [Kontopoulos et al. \(2008\)](#) present an ontology-based recommender system for e-course generation called PASER (Planner for the Automatic Synthesis of Educational

Resources) which can provide personalized curricula to the learner. Neri and Colombetti (2009) recommender system also uses domain ontology knowledge in retrieval of learning objects and generation of courses. Žitko et al. (2009) propose an ontology-based system for dynamical generation of questions as a test over previously formalized domain knowledge. Bahmani (2012) present an ontology-based recommendation algorithms that can personalize course and curriculum content for individual students. Pukkhem (2013) on the other hand propose an ontology-based semantic approach for learning object recommendation.

Moreover, Ruiz-Iniesta et al. (2014) introduce a knowledge-based recommender for educational resources of a computer science course. Similarly, Mota et al. (2014) propose a knowledge-based recommender system supported by an ontological modeling approach to assist educators in designing of teaching-learning activities. Shishehchi and Banihashem (2011) propose a learning content recommendation for Visual Basic.Net programming language which uses ontology to represent knowledge about learner, learning materials and domain in their recommender. Wang et al. (2007) present a personalized learning objects recommendation model (LORM) that uses ontology and a hybrid method of preference-based algorithm and correlation-based algorithm. Rani et al. (2014) propose a hybrid approach for semantic question answering based on ontology similarity and fuzzy logic. Nowakowski et al. (2014) propose a recommendation approach that makes extensive use of ontologies to formally represent learning context. The approach has been implemented and evaluated with the OP4L (Online Presence for Learning) prototype. Cobos et al. (2013) present an ontology-based hybrid recommender system named Recommendation System of Pedagogical Patterns (RSPP) which combines both content-based and collaborative filtering techniques. This system allows lecturers to define their best teaching strategies for use in the context of a specific class.

5.2 Analysis of results

This subsection presents the analysis of results of our survey from the 36 publications reviewed and analyzed in this study. The first part presents the classification of publications according to the year of publication. The second part is categorization of recommendation techniques while the last part is categorization of knowledge representation techniques, ontology type, ontology representation language, as well as types of recommended learning resources.

5.2.1 Classification of publications on ontology-based recommenders for e-learning

The relevant papers on ontology-based recommender systems for e-learning in this survey were classified according to the year of publication (Fig. 4). The purpose of classification was to give a clear picture on the distribution of publications for the last 10 years as well as summarize the achievements made in this area in terms of research output. A total of 36 journal paper publications relevant to ontology-based recommender systems for e-learning were analyzed and classified according to the year of publication. All the 36 papers classified in this review study were published between 2005 and 2014. A total of 26 of the papers were published between 2010 and 2014.

From Fig. 4, it is evident that there was significant growth in the number of publications on ontology-based recommendation for e-learning from 2005 to 2014. In the years 2005 and 2006, there was only one journal publication for each year relevant to this study followed by a steady increase between the years 2007–2010. The sudden surge in the year 2010 may have been as a result of evolutions in other related areas such as e-learning and web 2.0. A

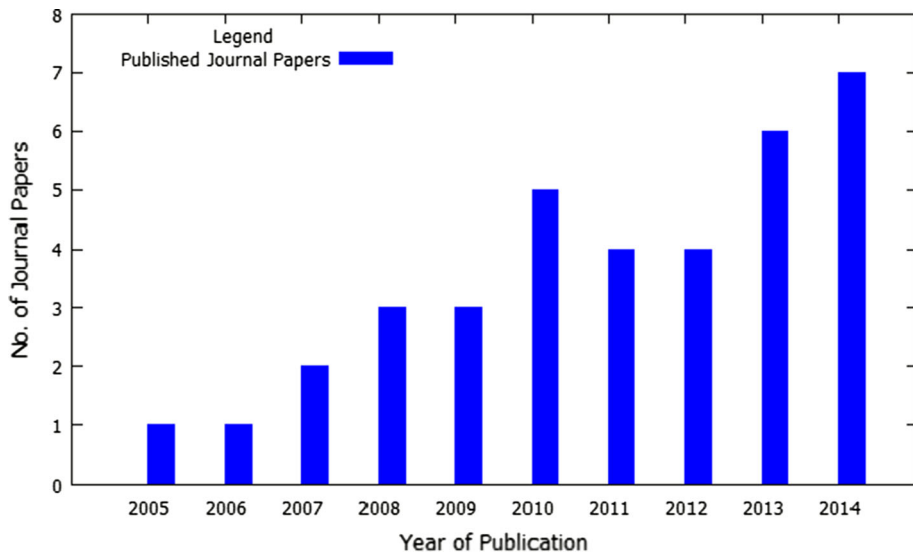


Fig. 4 Graph of distribution of journal papers according to year of publication. Total Number of Papers = 36

related study by [Park et al. \(2012\)](#) which classified recommender systems in general from 2001 to 2010 found out that in 2010, there was also a rapid increase in the number of research papers on recommender systems as research began to extend to other fields such as books, documents and music. From 2011 to 2014, the number of publications per year increased steadily again hitting the peak in 2014. This marks a significant growth in number of research publications in the area of ontology-based recommendation for e-learning from 2005 to 2014 (Fig. 4).

5.2.2 Recommendation technique(s) for ontology-based recommenders for e-learning

Table 1 tabulates the recommendation technique(s) used by each of the ontology-based (OB) recommenders for e-learning reviewed in this survey. It is evident from Table 1 that all the 36 recommender systems use either ontology-based or knowledge-based (KB) recommendation technique either in isolation or in combination with other recommendation techniques. All the 36 recommender systems in this review use ontology for knowledge representation (Table 2), hence they are classified as ontology-based recommenders. Knowledge-based and ontology-based techniques are commonly used in hybrid with other recommendation techniques. Other recommendation techniques used alongside ontology-based in this review include collaborative filtering (CF), content-based (CB), context-aware (CA), fuzzy-based (FB), and trust-aware based (TA) techniques. Other less commonly used techniques are semantic recommendation, rule-based, preference-based and correlation-based. For example, [Fraihat and Shambour \(2014\)](#) used a hybrid of ontology-based and semantic recommendation while [Zuhadar and Nasraoui \(2010\)](#) used a hybrid of content-based, knowledge-based and rule-based recommendation. [Wang et al. \(2007\)](#) combined preference-based and correlation-based approach in their ontology-based recommender for learning materials. Although ontology-based recommenders for e-learning use knowledge-based techniques for recommendation, in many cases they use a hybrid of knowledge-based technique with other recommendation

techniques as shown in Table 1. In this survey, 16 out of the 36 ontology-based recommenders used hybrid recommendation technique.

5.2.3 Knowledge representation in ontology-based recommenders for e-learning

Table 2 tabulates the techniques used for knowledge representation in the reviewed publications on ontology-based recommenders for e-learning. It is evident that all the 36 recommenders in this survey use ontology for knowledge representation alongside other techniques to support knowledge representation. Ontology-based recommenders for e-learning use ontology for knowledge representation as one of the knowledge representation techniques. Using ontology to model and represent knowledge about the learner, learning objects and domain knowledge is useful and most of the publications reviewed in this survey reported improved accuracy and performance of the recommenders. Among other techniques that support knowledge representation in this survey alongside ontology are web mining techniques, semantic reasoning techniques, spreading activation, context-awareness, learning style adaptation, singular value decomposition and sequencing rules. For instance, [Blanco-Fernández et al. \(2011\)](#), [Weng and Chang \(2008\)](#) and [Cantador et al. \(2008\)](#) use both ontology and spreading activation for knowledge representation. [Pukkhem \(2013\)](#) and [Dwivedi and Bharadwaj \(2013\)](#) used ontology with learning styles adaptation for knowledge representation (Table 2).

5.2.4 Ontology types and representation languages in ontology-based recommenders for e-learning

From Table 2, it is evident that majority (33 out of 36) ontology-based recommenders for e-learning make use of domain ontology. On the other hand, a few studies such as [Cobos et al. \(2013\)](#) used reference ontology, [Bahmani \(2012\)](#) used both generic and domain ontologies while [Vesin et al. \(2011\)](#) used both domain and task ontologies in their recommender systems.

Moreover, Table 2 shows that most ontologies in the surveyed ontology-based recommenders for e-learning were encoded with OWL representation language. Among the surveyed ontology-based recommenders, 19 out of 36 ontologies were encoded with OWL, while 4 out of 36 used RDF representation language. Some studies in this survey did not explicitly indicate the representation language used in their ontologies.

5.2.5 Learning resources recommended by ontology-based recommenders for e-learning

Table 2 further shows the different types of learning resources recommended by ontology-based recommender systems reviewed in this survey. It can be observed from Table 2 that the most common learning resources recommended by these ontology-based recommender systems for e-learning is learning materials. Other learning resources recommended by ontology-based recommender systems reviewed in this survey include questions, answers, courses, curricula, learning links, lectures and other learning objects. Table 2 summarizes the types of learning resources recommended by the ontology-based recommender systems for e-learning reviewed in our work.

5.3 Discussion

The journal papers reviewed, classified and categorized in this survey were definitely high quality, especially because most of them were downloaded from Science Citation Index (SCI)

Table 1 Categorization of recommendation techniques used in OB recommenders for e-learning

Author(s), recommender system and citations	Hybrid	CF	CB	KB/OB	CA	FB	TA	Others
Pukkhem (2014)—LORecommNet—1	—	—	—	X	—	—	—	—
Capuano et al. (2014)—IWT—8	X	X	—	X	—	—	—	—
Ruiz-Iniesta et al. (2014)—3	X	—	—	X	X	—	—	—
Rani et al. (2014)—9	X	—	—	X	—	X	—	—
Fraihat and Shambour (2014)—Semantic RS—0	X	—	—	X	—	—	—	X
Baseera (2014)—E-learning modules RS—0	—	—	—	X	—	—	—	—
Nowakowski et al. (2014)—OP4L—3	—	—	—	X	—	—	—	—
Cobos et al. (2013)—RSPP—27	X	X	X	X	—	—	—	—
Pukkhem (2013)—5	—	—	—	X	—	—	—	—
Cheng et al. (2013)—2	X	—	X	X	—	—	—	—
Zhang et al. (2013a)—2	X	X	—	X	—	—	—	—
Wang and Huang (2013)—E-material RS—7	—	X	—	X	—	—	—	—
Dwivedi and Bharadwaj (2013)—10	X	X	—	X	—	—	X	X
Shishechi et al. (2012)—E-learning RS—9	—	—	—	X	—	—	—	—
Vesin et al. (2012)—Protus 2.0—52	—	—	—	X	—	—	—	—
Ferreira-Satler et al. (2012)—23	X	—	—	X	X	—	—	—
Bahmani (2012)—PERCEPOLIS—2	X	—	—	X	X	—	—	—
Vesin et al. (2011)—Protus—13	—	—	—	X	—	—	—	—
Blanco-Fernández et al. (2011)—25	X	—	X	X	—	—	—	—
Huang et al. (2011a)—Semantic RS—0	—	—	—	X	—	—	—	—
Shishechi and Banihashem (2011)—2	—	—	—	X	—	—	—	—
Hsu et al. (2010)—Reading RS—46	—	—	—	X	—	—	—	—
Zhuhadar and Nasraoui (2010)—7 (HyperManyMedia)	X	—	X	X	—	—	—	X
Ciuciu and Tang (2010)—Virtual teacher—1	—	—	—	X	—	—	—	—
Yang (2010)—16	—	—	—	X	—	—	—	—

Table 1 continued

Author(s), recommender system and citations	Hybrid	CF	CB	KB/OB	CA	FB	TA	Others
Rey-lópez et al. (2010)—T-Learning 2.0–8	X	–	–	X	–	–	–	X
Biletskiy et al. (2009)—PSDLO—32	–	–	–	X	–	–	–	–
Žitko et al. (2009)—Questions RS—37	–	–	–	X	–	–	–	–
Neri and Colombetti (2009)—21	–	–	–	X	–	–	–	–
Weng and Chang (2008)—65	–	–	–	X	–	–	–	–
Kontopoulos et al. (2008)—PASER—72	–	–	–	X	–	–	–	–
Cantador et al. (2008)—78	X	–	X	X	–	–	–	–
Yu et al. (2007)—Content RS—64	–	–	–	X	–	–	–	–
Wang et al. (2007)—(LORM)—56	X	–	–	X	–	–	–	X
Mao et al. (2006)—DiLight—4	X	–	X	X	–	–	–	–
Shen and Shen (2005)—24	–	–	–	X	–	–	–	–

Table 2 Categorization by ontology type, representation language and recommended learning resources in ontology-based recommenders for e-learning

Author(s) and recommender system	Ontology type	Ontology representation language	Recommended learning resources
Pukkhem (2014) (LORecommendNet)	Domain ontology (learner and learning objects ontologies)	OWL with SWRL and Jess rules	Learning materials
Capuano et al. (2014)—IWT	Domain ontology (learner model and learning resource model ontologies)		Learning goals
Ruiz-Iniesta et al. (2014)	Domain ontology	OWL	Educational resources
Rani et al. (2014)	Domain ontology	OWL	Answers to questions
Fraihat and Shambour (2014) (Semantic RS)	Domain ontology	OWL	Learning materials
Baseera (2014) (E-learning modules RS)	Domain ontology		Course content
Nowakowski et al. (2014) (OP4L)	Domain ontology		Learning materials
Cobos et al. (2013)—RSPP	Reference ontology	OWL and RDF	Learning patterns
Pukkhem (2013)	Domain ontology (learner model and LO ontologies)	OWL with semantic mapping rules	Learning materials
Cheng et al. (2013)	Domain ontology	OWL	Learning materials
Zhang et al. (2013a)	Domain ontology		Nearest neighbors
Wang and Huang (2013) (e-material recommender)	Domain ontology		Learning materials
Dwivedi and Bharadwaj (2013) (E-learning RS)	Domain ontology		Learning materials
Shishechi et al. (2012) (e-learning recommender)	Domain ontology (learner and learning material ontologies)		Learning materials
Vesin et al. (2012)—Protus 2.0	Domain ontology (learner model, task and teaching strategy ontologies)	OWL with SWRL and Jess rules	Java programming content
Ferreira-Satler et al. (2012)	Domain ontology (fuzzy ontology)	OWL	Learning materials
Bahmani (2012)—PERCEPOLIS	Generic and domain ontology		Courses and curricula
Vesin et al. (2011)—Protus	Domain ontology and task ontology		Relevant learning links and actions
Blanco-Fernández et al. (2011) (Generic recommender)	Domain ontology (ontology of interest)	OWL & RDF	Digital TV content

Table 2 continued

Author(s) and recommender system	Ontology type	Ontology representation language	Recommended learning resources
Huang et al. (2011a) (semantic learning RS)	Domain ontology (learners, LOs and course ontologies)		Learning paths & content
Shishchichi and Banhashem (2011)	Domain ontology (learner and learning material ontologies)	OWL	Learning materials
Hsu et al. (2010) (English reading recommender)	Domain ontology		Reading material
Zhuahdar and Nasraoui (2010) (HyperManyMedia)	Domain ontology		Courses and lectures
Ciuciu and Tang (2010) (Virtual teacher)	Domain ontology (learning materials ontology)	OWL	Learning materials
Yang (2010)	Domain ontology		Information for scholars
Rey-lópez et al. (2010) (T-learning 2.0)	Domain ontology	OWL	TV programmes and LOs
Biletskiy et al. (2009)—PSDLO	Domain ontology (learner & LOs ontology)		Learning materials
Žitko et al. (2009) (Questions recommender)	Domain ontology	OWL	Tests/questions
Neri and Colombetti (2009)	Domain ontology	OWL-DL	Learning materials
Weng and Chang (2008)	Domain ontology		Research document
Kontopoulos et al. (2008) (PASER)	Domain ontology	RDF & XML	Personalized curricula
Cantador et al. (2008)	Domain ontology	OWL & RDF	Learning materials
Yu et al. (2007) (Content recommender)	Domain ontology (learner and learning content ontologies)	OWL	Learning materials
Wang et al. (2007)—LORM	Domain ontology		Learning materials
Mao et al. (2006)—DiLight	Domain ontology		Course materials
Shen and Shen (2005)	Domain ontology	OWL	Learning materials

indexed journals with an impact factor. Secondly, journal paper publications are more detailed in content and undergo rigorous review during the peer review process, hence they are of high quality. Figure 4 and Tables 1, 2 summarize the results of our review study. It is evident from the results of this survey that there was a significant increase in the number of publications on ontology-based recommendation for e-learning domain from 2005 to 2014. This is an indication of increasing research interest in this area of study. This significant growth and increase in research interest can be attributed to evolution and maturity of research in the wider area of recommender systems as well as adoption and acceptance of e-learning as an approach to teaching and learning by more institutions of higher learning (Tarus et al. 2015).

The results further reveal that ontology-based and knowledge-based recommendation techniques are widely used for recommendation of learning resources in e-learning environments. In addition, it was also evident that ontology-based recommenders in many cases use hybrid recommendation technique by combining knowledge-based techniques with other approaches such as collaborative filtering, content-based, context-aware, fuzzy-based and trust-based techniques (Table 1). Cobos et al. (2013) for instance used a hybrid of content-based and collaborative filtering approach at the same time using ontology for knowledge representation. Ferreira-Satler et al. (2012) and Bahmani (2012) both used a hybrid of ontology-based and context-aware recommendation techniques. Capuano et al. (2014) used a combination of collaborative filtering and ontology-based techniques whereas Zhuhadar and Nasraoui (2010) hybridized content-based and rule-based techniques for recommendation with the aid of ontology for knowledge representation. Table 1 illustrates the different recommendation techniques used in ontology-based recommenders for e-learning. Ontology-based recommendation is preferred and widely used for e-learning since it can alleviate the complexity of recommending learning resources to learners with different characteristics which has been cited by other researchers as a drawback to conventional recommendation techniques like collaborative filtering and content-based (Buder and Schwind 2012; Chen et al. 2014). Results from most of the publications reviewed showed improvement in performance and quality of recommendations by using ontology-based recommendation techniques in the e-learning domain (Pukkhem 2014; Yang 2010; Capuano et al. 2014).

Furthermore, results show that ontology is a popular approach for knowledge representation in ontology-based recommenders for e-learning. In this survey, all the 36 ontology-based recommenders use ontology for knowledge representation (Table 2). In the context of e-learning recommendation, ontology is used to represent knowledge about the learner, learning objects as well as the domain knowledge. Pukkhem (2013) points out that ontology also enables recommenders to interpret learner context and content features based on their ontologies. For instance, Shishehchi et al. (2012), Huang et al. (2011a), Pukkhem (2014) and Vesin et al. (2012) among others used ontology for knowledge representation in their recommender systems. Other techniques in this survey used for knowledge representation alongside ontology are singular value decomposition (SVD), data mining techniques, semantic reasoning techniques, fuzzy item response theory, context awareness, computational intelligence, ontology network analysis and sequencing rules.

Moreover, it is evident from this survey that most ontology-based recommenders for e-learning use domain ontologies. Other kinds of ontologies in use include reference ontology, task ontology and generic ontology. In addition, OWL representation language is widely used in encoding ontologies for ontology-based recommender systems. RDF representation language is also used in encoding ontologies for ontology-based recommenders though not widely used.

Lastly, it emerged that most of the ontology-based recommenders for e-learning recommend learning materials (Table 2). From the publications on ontology-based recommenders

that were reviewed and categorized, 20 out of 36 recommenders were learning material recommenders. This is expected mainly because most of the recommender systems within the e-learning domain target learners who are most of the time interested with learning materials. Other learning resources recommended by the reviewed systems include courses and curricula, learning paths, questions and answers. In the context of e-learning, ontology knowledge is used to map the learner to these learning resources. The recommender systems presented in [Fraihat and Shambour \(2014\)](#), [Nowakowski et al. \(2014\)](#) and [Wang and Huang \(2013\)](#) recommend learning materials to the learners. [Bahmani \(2012\)](#), [Zhuhadar and Nasraoui \(2010\)](#) and [Baseera \(2014\)](#) recommend courses whereas [Rani et al. \(2014\)](#) and [Žitko et al. \(2009\)](#) recommend questions/answers with the aid of ontology.

5.4 Limitations

Use of ontology for knowledge representation in ontology-based recommenders for e-learning has the potential to improve the quality of recommendations. This review study has revealed that there has been significant growth in research on ontology-based recommendation. However, there are still some challenges faced by researchers in this field.

First, creating ontologies is a difficult and time consuming process which requires skills in knowledge engineering. Furthermore, acquisition of ontology knowledge in the context of e-learning requires experts in this area. However, recent studies have proposed solutions to alleviate this difficulty. For instance, a user centred methodology called TORMES (Tutor-Oriented Recommendations Modeling for Educational Systems) has been proposed ([Santos and Boticario 2015](#); [Drachler et al. 2015](#)). This methodology can support educators in identifying recommendation opportunities in e-learning environments. The methodology can also be applied to elicit knowledge required to create ontologies in ontology-based recommender systems.

Secondly, evaluation of ontology-based recommenders for e-learning is another challenge experienced by many researchers in the area. This is partly due to scarcity of available standard public e-learning datasets for recommender systems evaluation. Unlike other domains like movies and books which have established public datasets for evaluation, public datasets for e-learning recommenders are scarce ([Manouselis et al. 2010](#)), hence making evaluation of ontology-based recommenders difficult. It has been suggested that where no real data is available in e-learning, data from other domains such as “MovieLens” dataset can be used ([Chughtai et al. 2014](#)) for initial testing but there will be need to use real world data during the final evaluation and comparison of performance with other similar systems.

Other limitations related to the scope of this review are

Choice of articles In this review, we considered only journal paper articles on ontology-based recommendation for e-learning in our classification and categorization of publications. Other articles such as conference papers and articles which are not subjected to peer review process were not included in the survey. In case we would have included papers from other articles, possibly the results would be different and more diverse. Journal papers were however preferred over other articles since they are more detailed and their peer review process is more rigorous, hence their ideas are more reliable and of high quality than other articles that are not peer reviewed.

Period of publications Our review considered only the papers published from 2005 to 2014. Perhaps, if a longer period would have been covered, the number of relevant papers would be slightly higher and might have some implications on the results. Nevertheless, research on ontology-based recommendation for e-learning domain is still new, hence a period of 10 years

can be considered sufficient for a comprehensive review. From Fig. 4, the first 2 years (2005 and 2006) had only one journal publication each implying that research in this area had not matured before the year 2005.

5.5 Future trends in ontology-based recommendation for e-learning

A number of studies on ontology-based recommendation for e-learning have been carried out in the last few years. However, there are still a number of challenges and areas of improvement that needs to be addressed in order to improve further the performance of the ontology-based recommenders for e-learning. In this sub-section, we present the possible future research direction towards improving the performance of ontology-based recommendation for e-learning.

Further research on ontology-based recommendation for e-learning An interesting trend in research on ontology-based recommendation for e-learning is the increasing interest for further research in this field. This renewed research interest is evident from the significant growth in the number of journal publications on this research field from 2005 to 2014 (Fig. 4). This increasing interest can be attributed to evolutions and success in research on recommender systems in general, the web and data mining technologies. This is also motivated by the fact that e-learning is gaining more acceptance in institutions of higher learning as an approach to teaching and learning (Tarus and Gichoya 2015). To further improve the quality and performance of ontology-based recommenders for e-learning, there is need for further research in this field.

Hybridization of recommendation techniques There is a clear trend towards hybridization of ontology-based recommendation for e-learning with other recommendation techniques such as collaborative filtering, content-based, social network-based, fuzzy-based, trust-based or context-aware based techniques. Hybridization of ontology-based recommendation with other techniques has the potential of improving the performance of the recommenders. In addition, hybridization of ontology-based recommendation with other advanced recommendation techniques helps to overcome the drawbacks associated with the conventional recommendation techniques such as cold-start and data sparsity problems. Such hybridization has also been suggested by other researchers. There is need for further research on hybridization of ontology-based recommendation for e-learning with other advanced recommendation techniques with a view to improving their effectiveness and performance.

Hybridization of knowledge structures Another interesting trend is on hybridization of ontology with other knowledge structures for knowledge representation in ontology-based recommendation for e-learning. Hybridization of knowledge structures has the potential to improve the accuracy and quality of recommendations. Though ontology is currently the most commonly used tool for knowledge representation in ontology-based recommenders for e-learning, there is a tendency to combine ontology knowledge with other knowledge structures to improve the quality of recommendations. Other knowledge structures include knowledge vectors (Rodrigues et al. 2012), case-based reasoning (Gutiérrez et al. 2012), social knowledge (Carrer-Neto et al. 2012) and constraint-based reasoning (Felfernig and Burke 2008). This trend goes hand in hand with the advancements in knowledge-based systems research.

Diversification of ontology knowledge sources Future ontology-based recommender systems for e-learning will likely rely on diverse sources of ontology knowledge both implicitly and explicitly. There is a wealthy of ontology knowledge available on the web that may be relevant to the learner and learning resources. Extraction of ontology knowledge both implicitly

and explicitly from diverse sources can improve the quality of ontology knowledge and by extension the quality of recommendations. This trend goes hand in hand with advancements in other fields such as web mining, semantic web and machine learning.

6 Conclusions

Recommender systems in e-learning play an important role in addressing the problem of information overload as well as aiding the learners to find relevant and useful learning resources from large volumes of online learning resources available on the internet. Whereas conventional recommender systems generally recommend items to the users based on ratings, ontology-based recommenders for e-learning on the other hand add a new dimension of ontology domain knowledge about the learner and learning resources into the recommendation process.

In this paper, we have presented a comprehensive review of ontology-based recommendation for e-learning. The contribution of our work in this survey is two-fold. First, we have summarized the research achievements in the class of ontology-based recommenders from 2005 to 2014 by classifying the papers according to the year of publication and categorizing techniques with a focus on recommendation techniques, knowledge representation, ontology types, ontology representation language, and recommended learning resources. Secondly, we have given an insight on the future trends on ontology-based recommendation for e-learning.

This survey has revealed that ontology-based recommendation in combination with other recommendation techniques are widely used for recommendation of e-learning resources. Furthermore, it is evident from the papers reviewed that aggregation of ontology domain knowledge into the recommendation process can improve the accuracy and quality of recommendations and at the same time help to overcome the drawbacks associated with the conventional recommenders such as cold-start and data sparsity problems. Results further show that ontology is a useful tool for knowledge representation in ontology-based recommenders for e-learning. Future research will focus more on hybridization of recommendation and knowledge representation techniques to improve the quality of recommendations in ontology-based recommendation for e-learning. We hope that this review study will widen the frontiers of knowledge and provide useful literature for researchers interested in advancing this field of research.

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Compliance with ethical standards

Conflict of interests The authors declare that they have no conflict of interest.

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